



Should a standing or seated reference posture be used when normalizing seated spine kinematics?

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ABSTRACT

Currently in the literature there is no consensus on which procedure for normalizing seated spine kinematics is most effective. The objective of this study was to examine the changes in the range of motion (ROM) of seated posture trials when expressed as a percent of maximum standing and seated ROM. A secondary purpose was to determine whether the typical maximum planar calibration movements (flexion, lateral-bend, and axial twist) elicited the respective maximum ROM values for each spine region versus postures with specific movement instruction. Thirteen male participants completed seven different movement trials. These consisted of the maximum planar movement trials, with the remaining four postures being combinations of specific lumbar and thoracic movements. Global and relative angles for the upper-thoracic, mid-thoracic, lower-thoracic, and lumbar regions were calculated and normalized to both a seated and standing reference posture. When normalizing both global and relative angles the standing reference appears optimal for flexion, twisting and lateral bend angles in all spine regions, with the exception of relative flexion angle in the mid-thoracic region. The maximum planar movement trials captured the greatest ROM for each global angle, relative lower-thoracic angle and relative lumbar flexion angle, but did not for all other relative angles in the upper-thoracic, mid-thoracic, and lumbar regions. If future researchers can only collect one reference posture these results recommend that a standing reference posture be collected for normalizing seated spine kinematics, although a seated reference posture should be collected if examining relative flexion angles at the mid-thoracic region.

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1. Introduction

A recent study by Bauman et al. (2011) found that the typical Canadian spends on average five hours seated every weekday. With the average individual spending a large portion of their workday seated, there has been an increase in research examining the effects of sitting on spinal postures. The literature suggests the occurrence, or reoccurrence, of low back pain can be provoked through sitting (Sjolie, 2004), with occupations requiring seated work for greater than 95% of the total shift being associated with low back pain (Hoogendoorn et al., 2000). As more studies quantify the spine's movement and postures while seated, such as Claus et al. (2009), it is important to consider individual differences when examining and interpreting the results, and the use of a suitable normalization procedure (i.e. as a percent of maximum range of motion (ROM)). It has been suggested that postures and movements be expressed in terms of maximum range in order to make comparison between

individuals and between groups (Edmondston et al., 2007b). To date, few studies have normalized lateral bend or axial twist postures in either the thoracic or lumbar spine. Furthermore, seated lumbar spine ROM during flexion is typically normalized to a maximum standing reference posture (Callaghan and Dunk, 2002; Callaghan and McGill, 2001; Dunk and Callaghan, 2010), and it remains unclear whether this is appropriate or if a maximum seated reference posture should be used (i.e. the maximum angles achieved during sitting and standing differ).

While seated lumbar spine postures are often normalized using maximum standing flexion, there is no consensus on the most effective way to report seated thoracic spine kinematic data. Typically, works investigating seated thoracic spine motion have reported their findings as angles in degrees (Edmondston et al., 2007a; Gilleard et al., 2002), with few studies reporting normalized thoracic angles. Strahan et al. (2011) examined the kinematics of certain rowing techniques by quantifying the lower thoracic and lumbar position during the rowing stroke, for all three axes, as the percent of the seated rowing positions' ROM. Recently Nairn et al. (2013), investigated the thoracic spine during upright sitting, slumped sitting, and seated maximum flexion when normalized to maximum forward flexion from both sitting and standing

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postures. The results found that in the upper-thoracic region, seated maximum flexion values were greater than 200% when normalized as a percent of maximum standing flexion (Nairn et al., 2013). This research raised the question of whether a standing reference posture is the most appropriate way to normalize seated thoracic spine kinematics. Additionally, while there is a foundation for normalizing flexion patterns, a gap in the literature remains regarding normalization of axial twist and lateral bend data, as studies by Gregersen and Lucas (1967), Preuss and Popovic (2010), and Willems et al. (1996) have presented multi-axis spine kinematics in degrees.

During analyses of seated spine kinematics it is important to consider whether the presentation of normalized data, either from a maximum standing reference posture or maximum seated posture, can alter the results. Therefore, the primary purpose of this study was to examine the changes in the maximum ROM of seated posture trials when expressed as a percent of maximum standing (%STAND) or a percent of maximum sitting (%SIT) for flexion, lateral bend, and axial twist motion. A secondary purpose was to determine whether the maximum ROM was occurring in the intended maximum planar movement trial as opposed to one of the four seated movement trials.

2. Methods

2.1. Data collection

The study consisted of 13 right-hand dominant male participants with no history of neck, back and shoulder pain for at least one year prior to collection. The mean $\pm$ SEM age, height, and weight for the study was 23.6 ± 0.6 years, 1.82 ± 0.02 m, and 84.3 ± 3.9 kg, respectively. Informed consent was obtained for each participant prior to data collection and the institution's Office of Research Ethics approved all protocols prior to the study's commencement.

Kinematic data were collected at 50 Hz using a seven camera three-dimensional motion capture system (Vicon MX, Vicon Systems Ltd., Oxford, UK) to track the position of 31 passive-reflective markers. As shown in Fig. 1, 24 of the reflective markers were placed on eight rigid plates (three markers per plate), which were attached to the skin over specific vertebrae (T₁, T₄, T₅, T₈, T₉, T₁₂, L₁, and the posterior–superior iliac spines (PSISs)) as per Nairn et al. (2013). The remaining markers were placed over C7, greater trochanters, and the iliac crests and anterior–superior iliac spines of the pelvis.

Participants were asked to perform seven different movements which were performed unilaterally to the right where appropriate (i.e. axial twist and lateral bend). Three of these movements were maximum planar movement trials: maximum flexion (MaxFlex), maximum axial twist (MaxTwist), and maximum lateral bend (MaxBend), which were completed both while sitting (-Sit) and standing (-Stand). The remaining four movements were performed while seated only and used as the comparison postures with focus on thoracic twist or flexion, with the lumbar spine being in a neutral or at a mid-range flexion posture. These combinations included: thoracic twist with a neutral lumbar (Twist–Neutral), thoracic twist with a flexed lumbar (Twist–Flex), thoracic flexion with a flexed lumbar (Flex–Flex), and thoracic flexion with a neutral lumbar (Flex–Neutral). Participants were asked to begin in an upright posture, move to the specified end posture in approximately four seconds, hold the posture for five seconds (hold-phase), then to return to their starting position in another four seconds. The timing was assisted by both a metronome and researcher cues. For the maximum planar movements, each participant was told to flex/bend/twist as much as possible without specific instruction on how to move the different spine regions. In addition to the overall trunk movement, specific to each posture, standing flexion required the knees to remain straight; all lateral bend trials had the ear bent to the shoulder; and for the twist trials the head/neck was rotated so the participant was looking over their right shoulder. The specific head movement instructions were given to facilitate the full spinal range of motion and were similar to those used by Willems et al. (1996). During the sitting trials, the chair height was adjusted so the knee and hip angles were at approximately 90°. Each movement was repeated for three trials consecutively and each block of three was presented in a random order.

2.2. Data processing

The data were filtered using a fourth-order dual-pass, low pass Butterworth filter, with a cut off frequency of 2.5 Hz, as determined through a residual analysis (Winter, 2009). The spine was partitioned into four regions based on the location of the rigid plates: upper-thoracic (T₁–T₄), mid-thoracic (T₅–T₈), lower-thoracic (T₉–T₁₂), and lumbar (L₁–PSIS) (Nairn et al., 2013) as shown in Fig. 1. Angles for each region were expressed as both relative and global angles, with a regional relative angle

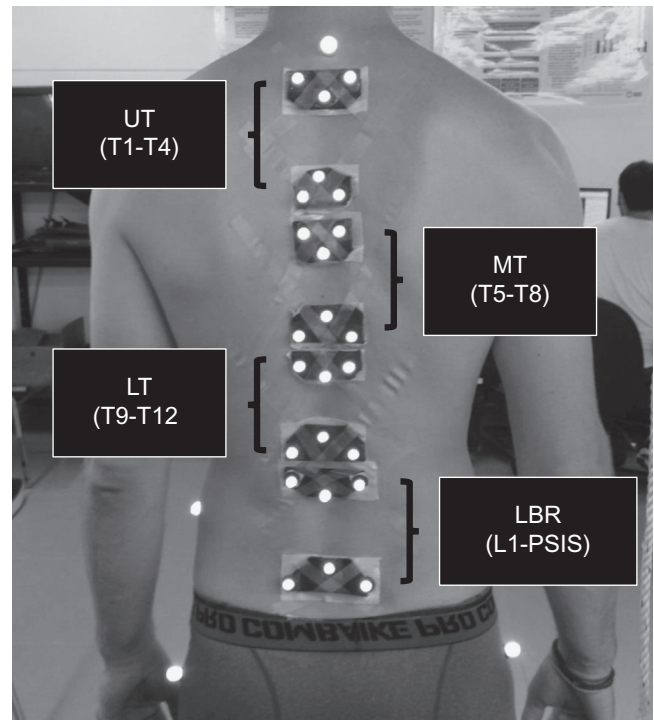


Fig. 1. Photo of the spine marker plate configuration and the respective spine segments generated from the plates. Segments include: upper-thoracic (UT), mid-thoracic (MT), lower-thoracic (LT), and lumbar (LBR).

being defined as the movement of a segment relative to the one below it, with the lumbar segment being relative to the pelvis. Global regional angles were taken as the particular segment with respect to the laboratory space. The three-dimensional rotation angles were calculated using an X–Y–Z (Flexion–Lateral Bend–Axial Twist) Cardan sequence. Given the mathematical nature of the analysis it is possible slight variations in results may have arisen based on the rotation sequence used. However, this order has been used previously during three-dimensional, multi-segmented trunk analyses of seated motion and follows orthopedic convention (Preuss and Popovic, 2010), and thus was deemed acceptable.

For each movement, the maximum ROM of each region was taken as the absolute difference between the maximum angle during the hold-phase of the posture and the participants' upright starting position. The upright starting position was determined for each participant and each region by averaging the first one second of data from all seated and standing trials separately, as trials began seconds before participants commenced their specified movements. For each participant a single starting position for sitting and one for standing was obtained for each of the four regions, in each of the three planes of movements. This resulted in 0° (0%Max) at each spine region being equivalent to upright. From each plane, the maximum angle of each region from any of the three repeats of the maximum planar movement trials (MaxFlex, MaxTwist, MaxBend) was recorded for each of the four regions, and this was done for both the maximum sitting and maximum standing trials separately. These values were then used for normalizing the regional angles from the specific seated posture movements to a percent of maximum ROM both while sitting (%SIT) and standing (%STAND).

2.3. Statistical analysis

The normalized ROM of each region (global and relative angles) for flexion, axial twist and lateral bend, were analyzed separately using 2×6 repeated measures ANOVAs, with the two factors being: reference posture (%SIT or %STAND) and Movement trial (Flex–Neutral, Flex–Flex, Twist–Neutral, Twist–Flex, Max–Sit, and Max–Stand). For each plane, all of the comparison movements as well as the specific seated and standing maximum planar movement trials for that plane were analyzed. For all analyses, $\alpha=0.05$ and all significant omnibus *F*-tests were further analyzed pairwise, and the *p*-value was adjusted using a Bonferroni correction. All statistical analyses were completed using a statistical software package (IBM SPSS Statistics, SPSS Inc., v.20.0, Chicago, IL).

3. Results

The main outcomes of this study were the recommendations made regarding the most appropriate reference posture for each

regional angle (relative and global), and the evaluation of the ability of the maximal planar trials to elicit a maximal angle (Table 1). Briefly, global angles were consistent across planes and regions for optimal reference posture and maximal trials capturing the largest ROM, whereas relative angles showed varying results on the type of reference posture and effectiveness of the maximal trials depending on the region. Summaries of the angular range of motion data for the global and relative angles are presented in Tables 2 and 3, respectively.

3.1. Global angles

The analysis of global angles revealed a significant interaction of reference posture and movement trial for all flexion and axial twist percentages ($F(5,60) > 12.00$, $p < 0.001$). Examining the maximum-trials (MaxFlex/Twist-Sit; MaxFlex/Twist-Stand), normalizing to %SIT as opposed to %STAND always showed significantly larger percentages, with differences ranging from 14.60% MaxFlex in the *upper-thoracic* region up to 191.96%MaxTwist in the *lumbar* region ($p < 0.003$) (Table 1). By resulting in a larger

percent, it is evident that the movement is capturing a smaller ROM. Conversely, lateral bend showed an effect of movement trial for all regions ($F(5,60) > 138.63$, $p < 0.001$), indicating no difference between %SIT and %STAND. Moreover, the maximum global angle was attained from their respective maximum planar movement trial for all spine regions ($p < 0.011$).

3.2. Relative angles

The *upper-thoracic* region showed an effect for movement trial that was found in all three planes ($F(5,60) > 8.72$, $p < 0.001$) with no main or interaction effects of reference posture ($F(5,60) < 1.92$, $p > 0.105$), indicating no difference between relative angles presented as a %SIT or %STAND (Table 1). However, the maximum planar movements did not appear to capture the greatest ROM for all three planes (Table 1, Fig. 2a).

Results for the relative *mid-thoracic* region showed an interaction between reference posture and movement trial for flexion, while only movement trial was found to be significant for twist and lateral bend ($F(5,60) > 7.37$, $p < 0.001$). Normalizing *mid-thoracic*

Table 1

Summary of the applicability of using a seated or standing reference posture, and whether the maximum-based trials provided the maximum angle for each spine region. The reporting of “Either” as the Suggested Reference Posture indicates that no statistical differences were found between values normalized to a standing reference posture versus a sitting reference posture.

Regional angle		Suggested reference posture (sit/stand/either)			Maximum angle from maximum trial? (yes/no)			If no, brief explanation		
		Flexion	Lateral bend	Axial twist	Flexion	Lateral bend	Axial twist	Flexion	Lateral bend	Axial twist
Global	Upper-thoracic	Stand	Either	Stand	Yes	Yes	Yes	–	–	–
	Mid-thoracic	Stand	Either	Stand	Yes	Yes	Yes	–	–	–
	Lower-thoracic	Stand	Either	Stand	Yes	Yes	Yes	–	–	–
	Lumbar	Stand	Either	Stand	Yes	Yes	Yes	–	–	–
Relative	Upper-thoracic	Either	Either	Either	No	No	No	Flexion trials resulted in greater maximum values than MaxFlex	Twist trials similar to MaxBend	Twist trials similar to MaxTwist
	Mid-thoracic	Sit	Either	Either	No	No	No	Flexion trials similar to MaxFlex	Twist trials similar to MaxBend	Some twist trials similar to MaxTwist
	Lower-thoracic	Stand	Either	Either	Yes	Yes	Yes	–	–	–
	Lumbar	Stand	Stand	Either	Yes	No	No	–	Some twist trials similar to MaxBend	Twist trials similar to MaxTwist

Table 2

Summary of mean $\pm$ SEM Global range of motion ($^{\circ}$) for each movement trial for the different spine segments. Seated postures include: thoracic flex with lumbar flex (Flex–Flex), thoracic twist with lumbar flex (Twist–Flex), thoracic flex with lumbar neutral (Flex–Neutral), thoracic twist with lumbar neutral (Twist–Neutral), and the maximum planar movements for both sitting and standing.

Flexion	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxFlex-Sit	MaxFlex-Stand
Upper-thoracic	48.1 (2.2)	30.5 (1.0)	27.5 (3.1)	4.3 (0.8)	101.4 (4.2)	121.1 (4.7)
Mid-thoracic	34.1 (3.5)	31.2 (2.0)	15.2 (3.4)	7.1 (1.2)	88.4 (8.5)	112.2 (10.5)
Lower-thoracic	23.1 (1.7)	24.9 (1.9)	5.2 (2.1)	3.7 (1.0)	78.8 (3.7)	112.2 (4.8)
Lumbar	10.8 (2.0)	11.5 (2.4)	1.6 (0.7)	2.3 (0.4)	54.2 (2.9)	76.7 (4.5)
Axial twist	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxTwist-Sit	MaxTwist-Stand
Upper-thoracic	2.9 (0.7)	37.7 (3.2)	2.1 (0.4)	38.5 (4.1)	56.0 (4.5)	93.3 (9.9)
Mid-thoracic	1.2 (0.4)	27.4 (3.6)	1.1 (0.4)	28.7 (4.1)	42.9 (4.3)	76.3 (7.9)
Lower-thoracic	1.7 (0.5)	13.5 (3.1)	1.3 (0.4)	12.0 (3.4)	25.1 (4.0)	59.9 (8.2)
Lumbar	1.5 (0.4)	7.6 (2.0)	1.0 (0.4)	5.7 (1.9)	16.2 (2.4)	49.5 (7.5)
Lateral bend	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxBend-Sit	MaxBend-Stand
Upper-thoracic	1.6 (0.3)	15.4 (2.6)	1.2 (0.4)	12.5 (2.2)	41.1 (2.1)	43.1 (2.2)
Mid-thoracic	1.8 (0.4)	3.1 (1.1)	1.2 (0.4)	1.7 (0.5)	46.6 (3.1)	45.9 (2.6)
Lower-thoracic	2.2 (1.6)	2.8 (0.5)	2.2 (0.6)	12.2 (3.4)	38.6 (3.4)	37.1 (1.8)
Lumbar	1.6 (0.3)	2.7 (0.3)	1.0 (0.2)	1.7 (0.4)	18.9 (2.7)	17.8 (1.1)

Table 3

Summary of mean $\pm$ SEM Relative range of motion ($^{\circ}$) for each movement trial for the different spine regions. Seated postures include: thoracic flex with lumbar flex (Flex–Flex), thoracic twist with lumbar flex (Twist–Flex), thoracic flex with lumbar neutral (Flex–Neutral), thoracic twist with lumbar neutral (Twist–Neutral), and the maximum planar movements for both sitting and standing.

Flexion	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxFlex-Sit	MaxFlex-Stand
Upper-thoracic	11.9 (0.8)	4.3 (0.6)	12.1 (0.9)	4.0 (0.7)	7.8 (1.1)	1.9 (1.8)
Mid-thoracic	14.6 (1.6)	12.5 (1.6)	13.1 (1.5)	7.5 (1.6)	17.0 (1.9)	12.6 (1.6)
Lower-thoracic	17.2 (1.6)	15.0 (1.7)	10.2 (2.1)	3.7 (0.8)	25.0 (1.4)	35.8 (2.1)
Lumbar	5.2 (0.5)	6.3 (0.7)	3.4 (0.6)	2.9 (0.4)	17.1 (1.4)	27.8 (2.8)
Axial twist	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxTwist-Sit	MaxTwist-Stand
Upper-thoracic	3.5 (0.6)	10.4 (1.5)	2.8 (0.5)	9.9 (1.7)	13.0 (2.3)	13.1 (2.0)
Mid-thoracic	3.1 (0.5)	10.3 (1.0)	3.2 (0.5)	13.6 (1.6)	15.6 (1.5)	15.5 (1.5)
Lower-thoracic	17.2 (1.6)	15.0 (1.7)	10.2 (2.1)	3.7 (0.8)	6.1 (1.0)	5.2 (1.3)
Lumbar	5.2 (0.5)	6.3 (0.7)	3.4 (0.6)	2.9 (0.4)	3.6 (0.6)	3.4 (0.4)
Lateral bend	Flex–Flex	Twist–Flex	Flex–Neutral	Twist–Neutral	MaxBend-Sit	MaxBend-Stand
Upper-thoracic	2.4 (0.2)	10.0 (1.3)	2.4 (0.3)	10.5 (1.6)	6.3 (0.5)	7.2 (0.8)
Mid-thoracic	2.1 (0.3)	6.5 (1.2)	2.5 (0.4)	8.1 (1.1)	11.3 (0.8)	10.4 (1.0)
Lower-thoracic	2.2 (1.6)	7.9 (1.7)	2.0 (2.1)	10.0 (0.8)	19.8 (1.4)	19.8 (1.3)
Lumbar	1.6 (0.5)	4.9 (0.7)	1.1 (0.6)	5.6 (0.4)	2.4 (0.4)	5.1 (1.0)

flexion angles as a %STAND resulted in a larger percentage than normalizing as a %SIT ($p < 0.001$) (Table 1, Fig. 2b). Comparing movement trials in the *mid-thoracic* region, the MaxFlex trials (both –Sit and –Stand) were not found to be different from the trials that specifically gave instructions for thoracic flexion (Flex–Neutral and Flex–Flex differences up to 19.47%MaxFlex, $p > 0.145$) (Fig. 2b).

Similarly the relative *lower-thoracic* region showed an interaction between reference posture and movement trial for flexion, and a main effect of movement trial for twist and lateral bend ($F(5,60) > 18.12$, $p < 0.001$). Normalizing these flexion angles as a %SIT resulted in larger percentages than as a %STAND, ($p < 0.001$) (Table 1, Fig. 2c). In all three planes the maximum angle came from the maximum ROM trials ($p < 0.04$) (Fig. 2c); the one exception was Twist–Neutral, which was $\sim 36.5\%$ MaxTwist less than MaxTwist–Stand, although this was approaching statistical significance ($p = 0.052$).

The *lumbar* region showed an interaction between reference posture and movement trial for flexion and lateral bend, whereas only movement trial was found to be significant for twist ($F(5,60) > 6.46$, $p < 0.001$). Both relative flexion and lateral bend angles presented as %SIT were found to be larger than the angle as %STAND (Table 1, Fig. 2d).

3.3. Reliability

The reliability of the maximum planar trials was assessed with intraclass correlation coefficients (ICCs) reporting Cronbach's alpha using a 2-way model. High reliability was shown across all measures (r range: 0.908–0.995) with the exception of one variable (relative *upper-thoracic* sitting lateral bend, $r = 0.645$).

4. Discussion

This study was the first to report normalized thoracic angle data in all three axes of motion, and to compare the effects of seated and standing reference postures on seated spine angle normalization (Table 1). It appears sufficient for all spine regions to normalize the global flexion and twisting angles to a percent of maximum standing, while either a standing or sitting reference posture is adequate for global lateral bend angles. However, the decision on whether a standing or sitting reference posture should be used for normalizing relative angles must be approached for each level of the spine and each direction individually. Flexion was the axis where the choice of reference posture varied the most between regions. Again briefly, for the *lumbar*, *upper-thoracic* and *lower-thoracic* regions, relative flexion angles should be normalized to a standing reference posture, as

angles as a %SIT were always larger or no different than angles as %STAND (Fig. 2a, c, d). The *mid-thoracic* was the only region where relative angles as %SIT were always smaller than the same angle as %STAND, suggesting a larger ROM while sitting (Fig. 2b). Relative lateral bend and twisting angles for all regions can be normalized as a percent of maximum standing, as the MaxBend trials expressed as %SIT were always larger or no different in comparison to %STAND. Ideally both reference postures would be collected, regardless of plane or region, with the largest angle from either reference posture being used for normalizing. If it is not feasible to collect both reference postures, due to time constraints or special population considerations, it appears acceptable to normalize relative three-dimensional spine kinematics to a standing reference posture, however a seated reference posture is recommended when examining relative flexion at the *mid-thoracic* region.

When normalizing kinematic data it is crucial to determine if the maximum movement trials are resulting in the largest angle. Across the four spine regions, the maximum movement standing trials produced global flexion, lateral bend, and twisting angles significantly greater than or equal to the maximum angle achieved during the sitting maximum movement trials (Table 1). While the maximum planar movement trials resulted in reference postures with the absolute largest angle when examining global angles, this was not the case when examining relative angles. With regards to the *lumbar* region, the MaxFlex–Stand trial resulted in greater relative flexion angles than all other trials (Fig. 2d); whereas in twisting and lateral bend, the comparison trials which called for twisting resulted in relative angles similar to those seen in both MaxBend and MaxTwist trials. For the *lower-thoracic* region, MaxFlex–Stand resulted in relative percentages greater than all other trials, while no difference was found between both MaxTwist, and MaxBend trials done while sitting compared to standing, although they were found to be significantly greater than all other movements. Examining the *upper-thoracic* and *mid-thoracic* regions, the maximum movement trials did not produce a ROM that was found to be significantly greater than all comparison movements. As comparison trials were able to elicit a greater magnitude of motion than the standard, or traditional, maximum planar movements, future research should investigate which sitting and standing postures result in the greatest absolute ROM, looking at each segment and plane of motion individually. As individual differences are present when examining the entire spine, as well as each spine region's ROM, it may be necessary to have participants complete a variety of movements attempting to capture their true maximum, with the trial resulting in largest ROM for each segment, being used for normalizing.

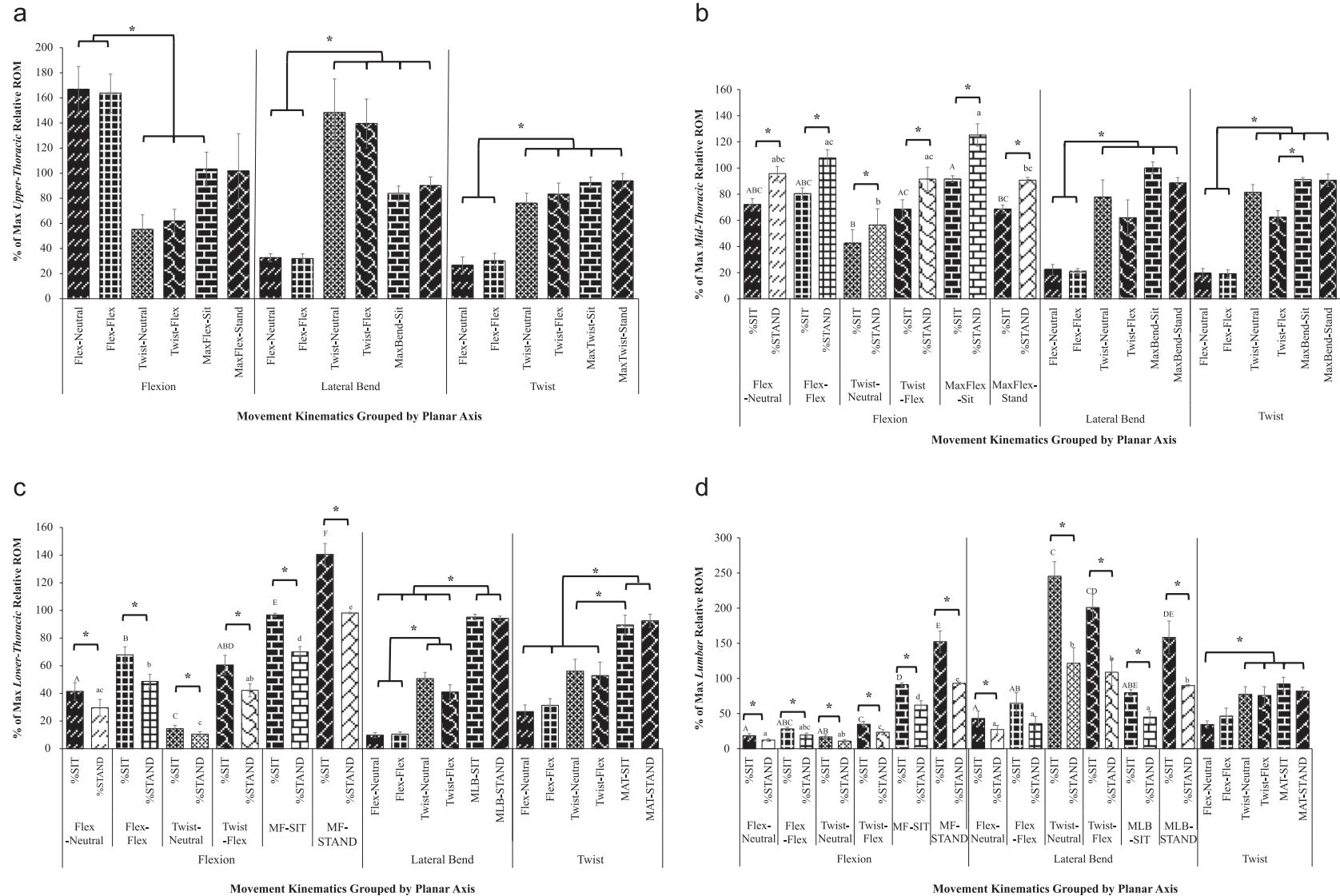


Fig. 2. Mean $\pm$ SEM normalized relative range of motion (ROM) from each plane is shown, with the respective maximum planar trials, for: (a) upper-thoracic, (b) mid-thoracic, (c) lower-thoracic, and (d) lumbar regions. When appropriate, different upper-case letters represent significant differences between %SIT postures whereas different lower-case letters represent significant differences between %STAND postures. Segments with no main or interaction effects of reference postures were collapsed across reference posture. In all cases *** denotes a significant difference between corresponding trials. $\alpha=0.05$ was used for all comparisons.

The angle data found in this study were comparable to those previously published in the literature by [Torén \(2001\)](#), [Doriot and Wang \(2006\)](#), [Willems et al. \(1996\)](#) and [Edmondston et al. \(2007a\)](#). For example, [Willems et al. \(1996\)](#) quantified angular ROM (degrees $\pm$ SD) relative to the pelvis segment for the *upper*-, *mid*-, and *lower-thoracic* during seated maximum planar movement trials and reported values of $7.8 \pm 5.0^\circ$ *upper-thoracic* flexion, $7.7 \pm 2.3^\circ$ *mid-thoracic* lateral bend and $8.3 \pm 6.1^\circ$ *lower-thoracic* twist. In the present study, the segmental angles relative to the pelvis were similar in range, with $7.0 \pm 5.0^\circ$ flexion, $11.3 \pm 2.8^\circ$ lateral bend and $11.1 \pm 5.9^\circ$ twist. The accompanying thoracic spine lateral bend coupling motion during seated maximum twisting, as well as the maximum thoracic twisting angles themselves ([Table 2](#)), were similar to those observed at the T_6 level by [Edmondston et al. \(2007a\)](#) of 41.8° twist and 3.3° lateral bend in a neutral position, as well as 33.5° twist and 7.1° lateral bend in a flexed position. The maximum lumbar flexion and twisting angles presently observed during MaxFlex-Sit, of $\sim 17^\circ$ ([Table 3](#)), and MaxTwist-Sit, $\sim 16^\circ$ ([Table 2](#)) respectively, were similar to the maximum angles observed during the maximum planar movement trials of [Doriot and Wang \(2006\)](#), $\sim 24^\circ$ lumbar flexion (relative), and [Torén \(2001\)](#), $\sim 22^\circ$ (global) lumbar twist. Therefore, the results of this study were comparable to those previously reported in the literature on seated spine kinematics.

The additional external support provided by a chair while sitting, and not just the position the spine adopts while seated, could partially explain the results of this study. While it was not observed during collection, to explain the lumbar twist data it was speculated that some participants used the chair and their feet as leverage when they attempted to maximally rotate. This method of producing “anchored” motion may have resulted in the observed twisting of the *lumbar* region in the opposite direction to the intended movement and could be responsible for the large lateral bend angles that occurred during twisting trials, as found in the Twist–Neutral and Twist–Flex trials. Of the thirteen participants, three participants may have used this anchoring technique during the comparison trials involving twist (Twist–Flex, Twist–Neutral), and one during the MaxTwist-Sit trials (produced global and relative twist angles to the left, noted during data analysis). Regardless, the magnitude of the movement for these participants was within the range of the other ten participants whose spines rotated in the intended direction of the movement. As a result of this, the absolute maximum angle achieved during the movement was used as the maximum angle, regardless of direction. The participant's global angle range over the three MaxTwist-Sit trials averaged to $\sim 5.34^\circ$, showing how little motion actually occurs at the lumbar region during seated twisting. The chair also seemed to affect relative flexion of the *upper-thoracic* angle, as the Flex–Neutral and Flex–Flex trials resulted in percentages greater than 175%STAND, similar to the finding of [Nairn et al. \(2013\)](#). This could be a result of the difficulty in maintaining balance when attempting to fully flex both the lumbar and thoracic spine at the same time while standing, which is less evident while sitting due to the support of the chair. As a result the overall largest movement from each region was taken as the ROM, in order to counteract any influence the chair may have had on the direction of the motion.

Similar to any study requiring motion capture, limitations were present in the procedure. The use of surface markers in motion analysis is a common concern as skin motion artefact can occur. As such, care was taken to ensure the rigid plates were adhered as best as possible and were constantly monitored for any gross-movement artefacts. Although our spine marker setup has been previously reported in the literature ([Nairn et al., 2013](#)) and was adapted from [Preuss and Popovic \(2010\)](#), it is possible that factors such as muscle contractions and marker configuration could have had an effect on the recorded movement. Sex has been identified as a factor that can

affect an individual's spine ROM ([Sullivan et al., 1994](#)), however these differences may be accounted for through the use of the proper normalization technique. Based on the work by [Van Herp et al. \(2000\)](#) and [Edmondston et al. \(2007a\)](#), it is expected that the maximal ROM elicited in a female population would also not be confined to the maximum planar trials as observed for the male participants. Unfortunately, due to the interference between the required marker placement configuration and the required female attire, only a male population could be used in this study. Also, it is possible that other variations of combined movement patterns not included in this study could produce a larger maximum than presented; and it is possible that an external device such as a bar could help facilitate the motion ([Johnson et al., 2012](#)). Considering this study was the first attempt to investigate the difference in using seated or standing normalization postures on seated movement trials, the aforementioned limitations were considered acceptable; and future work will be able to expand on the current findings.

5. Conclusion

This study illustrated the need to consider both the reference posture and type of trial used when normalizing seated spine kinematics across multiple levels of the spine. When investigating global angles in any direction it appears adequate to normalize angles as a percent of maximum ROM while standing. Regarding relative angles, a standing reference posture appears acceptable for normalizing spine kinematics, with the exception of the *mid-thoracic* region. Examining the different movement trials, global angles across all spine regions and planes indicated that the standard maximum planar movements consistently resulted in the largest global rotations compared to the comparison trials. Conversely, the standard maximum planar movement trials only resulted in the largest relative angle for the *lower-thoracic* angles and relative flexion in the *lumbar* spine. These findings highlighted the importance of considering not only the region of the spine, but how the data will be presented (as a global or relative angle), when normalizing seated spine kinematics. In conclusion, the ideal normalization procedure recommended for seated spine kinematics is to collect both seated and standing reference postures, and use the overall maximum angle attained for each spine segment. If it is only possible to record and/or have participants perform one reference posture, a standing reference posture would suffice, but caution should be exercised when interpreting the relative *mid-thoracic* flexion angles.

Conflicts of interest

The authors declare no conflict of interest.

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